COMPARATIVE EVALUATION OF VaR MODELS:

PARAMETRIC & HYBRID APPROACH

20163 — Risk Management and Value in Banking and Insurance

Group 15

Security 7 & Security 10

Parametric Approach (A)

Hybrid Approach (C)

Group 15 – Risk Management and Value in Banking and Insurance

MSc in Finance — Academic Year 2025/2026

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Assignment Overview

Security

Security 7

Fixed-Rate Semi-Annual

Coupon € 200

Face Value € 10,000

Maturity 31 Dec 2026

Security 10

Irregular coupon security

Coupon € 150

Face Value € 5,000

Maturity 30 Jun 2026

Risk factors

The risk factors used in the analysis are the log returns on ZCB prices.

Project Objectives

The purpose of this project is threefold:

- to implement the computation of Value-at-Risk (VaR) over a one-week horizon using a factor-based approach;
- to understand and apply cash flow mapping techniques (**clumping**), transforming complex bond cash flows into exposures to a limited set of yield curve nodes;
- to evaluate, through empirical analysis, how different modelling assumptions and **methodologies** influence VaR estimates.

The project concludes with a **comparative analysis** of the parametric and hybrid approaches and their implications for VaR estimation.

Methodologies applied

Clumping

Clumping maps bond cash flows onto a limited set of yield curve **nodes**, preserving their present value and interest rate sensitivity through **virtual cash flow** decomposition.

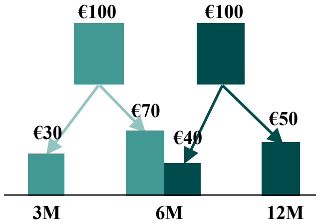
Parametric approach

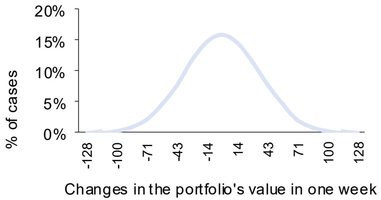
The **parametric approach** estimates risk by assuming a **normal distribution** for risk factors and using sensitivities and variance-covariance matrix to approximate how their changes affect the portfolio's value.

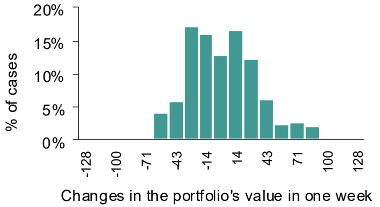
Hybrid approach

The hybrid approach combines historical and parametric modelling, applying exponentially weighted observations to capture time-varying volatility and estimate VaR through a weighted **percentile distribution**.

Expected results







Portfolio Set up – Clumping

Cash Flow Clumping

The clumping technique allows to identify, for each real cash flow, two virtual cash flows such that they are **identical in terms of sign, fair value, and risk (modified duration)**. The following system of equations is applied to determine the two cash flows.

$$\begin{cases} MV_s = MV_t + MV_{t+1} \\ MD_s = MD_t \dfrac{MV_t}{MV_s} + MD_{t+1} \dfrac{MV_{t+1}}{MV_s} \end{cases} \qquad MV_t = \frac{MD_{t+1} - MD_s}{MD_{t+1} - MD_t} MV_s \qquad MV_{t+1} = MV_s - MV_t$$

Step 1 – Compute of the **interpolated rate** for each cashflow date, using the interpolation formula

$$r_s = r_t + (r_{t+1} - r_t) \times \frac{(s-t)}{(t+1-t)}$$

Step 2 – Compute the **Modified Duration** for each cashflow to be clumped and for each of the adjacent nodes

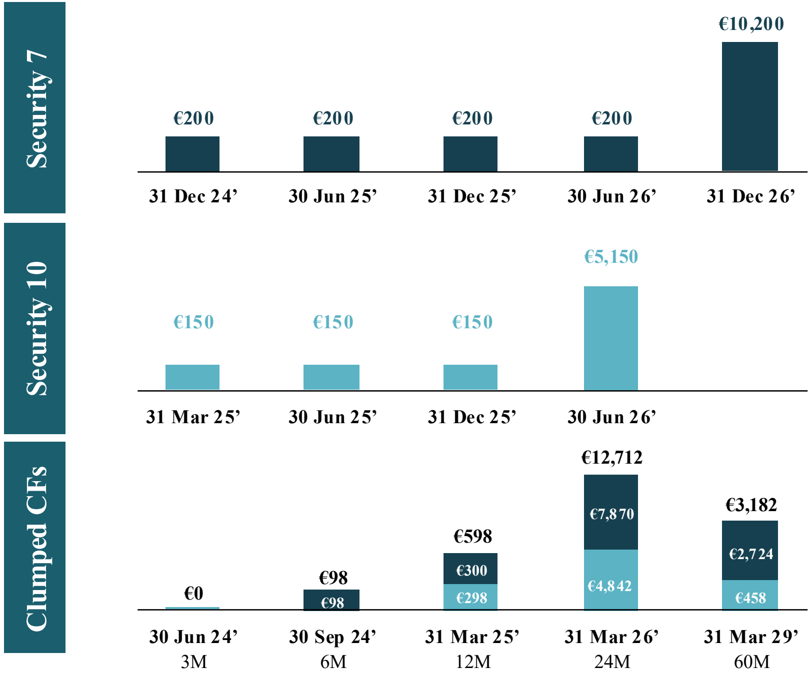
Step 3 – Find the **Market Value** of the real cash flows to clump

$$MV_s = \frac{CF_s}{(1+r_s)^s}$$

Step 4 – Apply the **system of equations** above and find the two market values (MV) of the virtual cash flows

Step 5 – Find the **associated Face Values** capitalizing until maturity

Example on 31 March 24' Evaluation date



Our initial portfolio had 6 different maturities. Thank to clumping we managed to map them to the 5 related to our term structure.

Parametric Approach — Methodology

The parametric approach estimates portfolio risk by assuming normally distributed risk factor changes and approximating portfolio value variations through a linear (first-order) expansion. The covariance matrix (Σ) and portfolio sensitivities (δ) are estimated and then combined to obtain the volatility of portfolio value changes ($\sigma_{\Delta V}$).

1 Portfolio representation

Bond cash flows are mapped onto a limited set of yield curve nodes (clumping), allowing the portfolio to be expressed as a combination of zero-coupon exposures.

$$\boldsymbol{V(P) = \sum_i CF_i \times P^{(i)}}$$

2 Computation of δ

Portfolio sensitivities are derived using a first-order approximation, where each delta corresponds to the market value exposure to the relevant risk factor.

$$\boldsymbol{\delta^{(i)} \cong \frac{\partial V}{\partial P^{(i)}} \times P^{(i)}}$$

3 Change in portfolio value

Based on a first-order Taylor expansion, the change in portfolio value is approximated as a linear function of risk factor returns:

$$\boldsymbol{\Delta V \approx \delta' \times r_{ln_w}}$$

4 Distribution of portfolio returns

Under the assumptions of zero-mean ("$\mu = 0$"), normally distributed risk factors and a correctly estimated covariance matrix ($\hat{\Sigma}$), portfolio value changes follow:

$$\boldsymbol{r_{ln_w} \sim N(0, \Sigma) \Rightarrow \Delta V \sim N(\delta' \times 0, \delta' \times \Sigma \times \delta)}$$

5 VaR estimation

VaR is computed within the delta-normal framework as:

$$\boldsymbol{VaR_p = z_p \times \sigma_{\Delta V}}$$

Σ and δ for the 31 Mar 24' evaluation date

2

$$\delta = \begin{bmatrix} - \\ 96 \\ 578 \\ 11,982 \\ 2,776 \end{bmatrix}$$

4

$$\hat{\Sigma} = \begin{bmatrix} 0.000001\% & \cdots & \cdots \\ 0.000001\% & \ddots & \vdots \\ \vdots & \cdots & 0.002198\% \end{bmatrix}$$

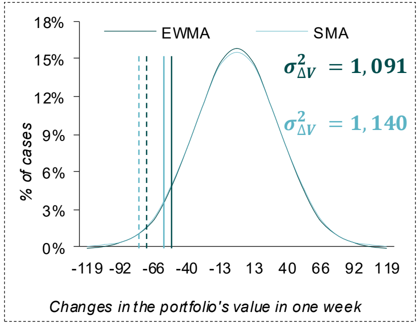
Note
For simplicity, only selected elements are shown; the full covariance matrix is 5×5 and the sensitivity vector has dimension 5×1.
Since the EWMA decay factor may excessively downweight past observations, the variance–covariance matrix is also estimated using a simple moving average. This allows us to assess potential weaknesses of the EWMA approach.

Main Pros & Cons

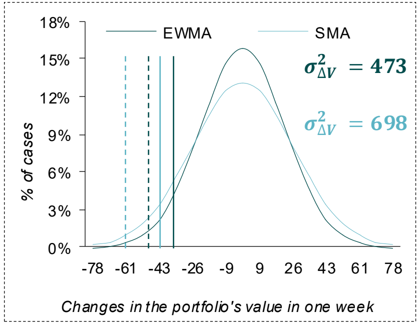
Pros	Cons
Computationally efficient	Normality assumption of risk factors
Easy to implement	Linear approximation
Applicable under mild conditions	Stable distribution of market factor returns
No need for full valuation models	Explicit volatility and correlation estimates

Findings Parametric approach

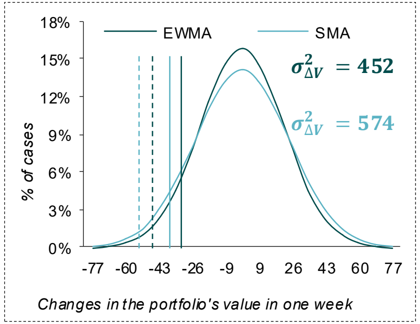
31 March 2024



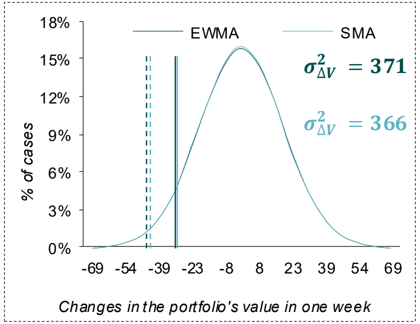
30 June 2024



30 September 2024



31 December 2024



99% CI EWMA | 95% CI EWMA | 99% CI SMA | 95% CI SMA

Considerations

The parametric approach highlights a clear **decline** in estimated portfolio risk across the four evaluation dates driven by the reduction in the estimated var-cov matrix over time. A key aspect is the comparison between the results using EWMA and SMA.

- The **SMA-based VaR** is consistently higher than the EWMA-based one, particularly in the earlier periods, due to the equal weighting of all past observations, hence capturing the higher volatility of observations which are far from the evaluation date.
- By contrast, the **EWMA specification** (λ = 0.94) places greater emphasis on recent observations, leading to a faster adjustment to changing market conditions.

As a result, while both approaches capture the same downward trend in risk, EWMA produces **more responsive estimates**, whereas SMA remains more influenced by past high volatility and therefore yields more conservative VaR levels.

Method	EWMA		SMA	
Evaluation date	95%	99%	95%	99%
31 Mar 2024	54.32	76.83	55.54	78.55
30 Jun 2024	35.76	50.57	43.45	61.45
30 Sep 2024	34.97	49.46	39.42	55.76
31 Dec 2024	31.69	44.82	31.48	44.53

Note: (1) SMA stands for Simple Moving Average

Hybrid Approach — Methodology

Hybrid approach consists in generating a potential distribution of the future value of discount factors to which our portfolio is mapped. The future prices of discount factors are obtained from past returns. Specifically, since the BRW approach, future prices obtained with returns more distant in time, will have a lower weight. Thanks to the BRW approach, discount factors which are more recent in time will have a stronger impact on the future value of the Portfolio.

Date	P_{ZCB3M}	P_{ZCB6M}	P_{ZCB12M}	P_{ZCB24M}	P_{ZCB60M}	ΔV	λ^i	Frequency
03/04/2023	0.99	0.98	0.97	0.94	0.87	9.60	0.040	0.27%
10/04/2023	0.99	0.98	0.97	0.94	0.87	-47.49	0.043	0.28%
17/04/2023	0.99	0.98	0.97	0.94	0.87	-49.46	0.045	0.30%
24/04/2023	0.99	0.98	0.97	0.94	0.88	24.76	0.048	0.32%
…	…	…	…	…	…	…	…	…
11/03/2024	0.99	0.98	0.97	0.94	0.87	2.10	0.831	5.52%
18/03/2024	0.99	0.98	0.97	0.94	0.87	-12.39	0.884	5.88%
25/03/2024	0.99	0.98	0.97	0.94	0.87	17.24	0.940	6.25%

Main Pros & Cons

Pros	Cons
No need for risk factor distribution assumptions	Requires a large historical sample
Full valuation	Computationally complex
Preserves past correlation and volatility estimates	Requires a pricing model for each position
More responsive to recent data	Very sensitive to the lambda assumption

1 Risk factor dynamics

Historical weekly (log) changes in zero-coupon bond prices are used to generate future scenarios. These changes are applied to current discount factors to obtain simulated future values.

$$P_{t+1}^{(i)} = P_t^{(i)} \times e^{r_{\ln w}^{(i)}}$$

2 Portfolio revaluation

For each scenario, the portfolio is fully revalued using updated discount factors obtained from simulated price changes.

$$V_{t+1} = \sum_i CF_i \times P_{t+1}^{(i)}$$

3 Computation of portfolio changes

Portfolio value changes are computed as the difference between simulated and current values, generating a distribution of profit and loss outcomes.

$$\Delta V = V_{t+1} - V_t$$

4 Weighting scheme

Observations are weighted using an exponential decay factor (EWMA), assigning greater importance to recent market movements and capturing time-varying volatility.

5 VaR estimation

VaR is obtained using the weighted empirical distribution of portfolio changes:

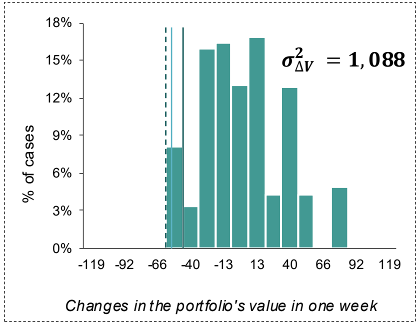
$$Var_p = \left|\Delta V_p - \mu_{\Delta V}\right| \quad \text{with} \quad \Delta V_p = \max\{x \mid \text{Prob}(\Delta V < x) \le p\}$$

An interpolation formula has been implemented to refine the VaR:

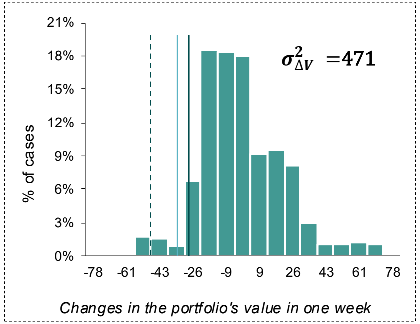
$$Var_p = \left| x_1 \times \frac{\alpha_2 - p}{\alpha_2 - \alpha_1} + x_2 \times \frac{p - \alpha_1}{\alpha_2 - \alpha_1} - \mu_{\Delta V} \right|$$

Findings Hybrid approach

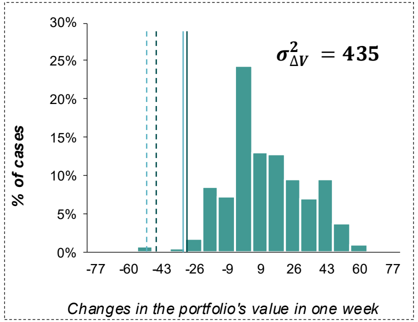
31 March 2024



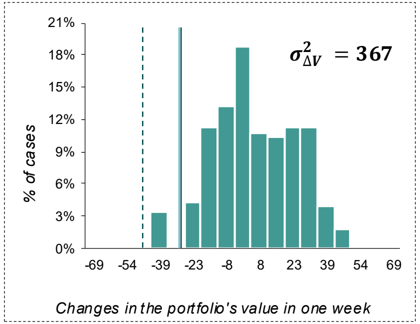
30 June 2024



30 September 2024



31 December 2024



99% CI Plain Vanilla **95% CI Plain Vanilla** **99% CI Interpolated** **95% CI Interpolated**

Considerations

Across the four evaluation dates, the hybrid approach shows a **clear reduction** in portfolio risk, reflected in the progressive narrowing of the loss distributions.

- In March 2024, the distribution is more dispersed and exhibits heavier tails, indicating higher tail risk, with a 99% VaR of about **60**. This reflects a more volatile market in the time window under analysis in the first period.
- Over time, the distribution becomes more concentrated around the mean, with **thinner tails** and **lower dispersion**, signaling a gradual decline in volatility.
- By December 2024, the distribution is more peaked and stable, with a lower 99% VaR of around **47**, indicating a significant reduction in extreme losses.

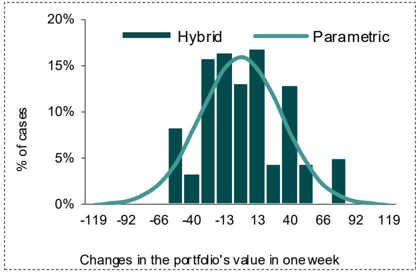
Overall, the hybrid approach captures a smooth adjustment of risk, driven by the increasing weight of more recent, lower-volatility observations.

Method	Plain vanilla VaR		Interpolated VaR	
Evaluation date	95%	99%	95%	99%
31 Mar 2024	51.38	60.07	57.30	n.m.[1]
30 Jun 2024	28.40	53.01	30.16	n.m.[1]
30 Sep 2024	29.43	48.80	30.92	54.83
31 Dec 2024	27.19	47.09	27.70	n.m.[1]

Note: (1) Interpolated VaR was in certain cases non measurable as the cumulative frequency of the lowest ΔV was greater than 1%

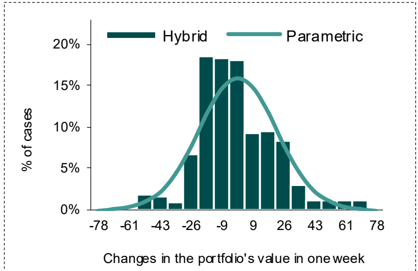
Comparison

31 March 2024



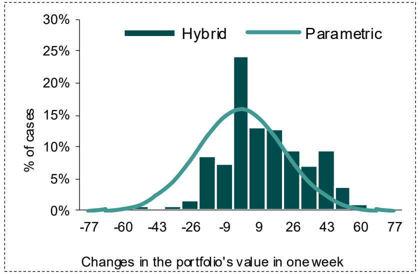
Hybrid approach data		
Mean	**Skewness**	**Exc. Kurt.**
1.60	-2.66	-0.32

30 June 2024



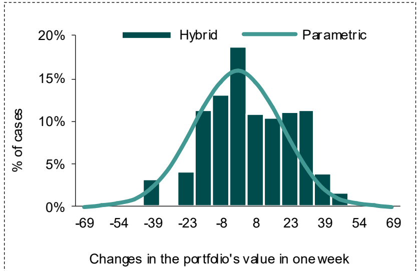
Hybrid approach data		
Mean	**Skewness**	**Exc. Kurt.**
-0.02	-2.44	1.13

30 September 2024



Hybrid approach data		
Mean	**Skewness**	**Exc. Kurt.**
11.60	-2.83	-0.44

31 December 2024



Hybrid approach data		
Mean	**Skewness**	**Exc. Kurt.**
5.00	-3.10	-0.28

Conclusions

In the graphs above, the parametric distribution is estimated using the EWMA approach, which yields variance levels comparable to those obtained under the hybrid method. This alignment allows us to isolate distributional effects and focus on the normality assumption rather than differences in volatility estimation.

Across the four reference dates, the **hybrid approach** shows clear deviations from normality, exhibiting noticeable **asymmetry** with consistently **negative skewness**, indicating a predominance of downside risk. This feature is not captured by the parametric approach, which imposes symmetry by construction.

Differences also emerge in **tail behavior**: while the parametric distribution smooths the tails, the hybrid approach reflects the empirical distribution more accurately, especially during periods of higher market stress where extreme observations are more relevant.

Regarding the zero-mean assumption, it appears to be broadly **reasonable**, as empirical means are small relative to dispersion, although slight deviations are present.

Overall, the results suggest that while the **zero-mean assumption on log returns is acceptable**, the **normality assumption is not fully supported by the data**. Consistently with this evidence, at the 95% confidence level the parametric VaR is generally higher, whereas at the 99% level the hybrid VaR tends to be larger, as there are some <u>extremely negative</u> values, driving down the skewness, which have a <u>strong impact at the 99% level, but have a limited impact on the 95% level.</u>

THANK YOU FOR YOUR ATTENTION !

Group 15

Chiara Sacconaghi

Raffaele Perillo

Ludovico Raspanti

Lorenzo Passioni

Christian Moresco